

Hai Bao

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Professional Experience

Sr. Data Scientist | Morningstar Sustainalytics | Jan 2023 - Mar 2026

- Streamlined data ingestion workflows to process 400,000+ news articles daily by engineering an automated ETL process using Python, reducing manual data handling and improving analysis turnaround.
- Optimized business support operations by integrating a GPT-powered LLM solution (OpenAI API), successfully reducing daily manual research tasks by 25% (1,000 articles) for the ESG Research team.
- Enhanced management's decision-making capabilities by engineering a Python-based Plotly dashboard (open source alternative to Power BI/Tableau) to track ESG Research workload distribution, enabling data-driven insights into team productivity and bottleneck identification.
- Helped Business Analysts with explaining AI model performance metric concepts such as Accuracy, Precision, Recall to the Senior Leadership Team.

Data Scientist | Morningstar Sustainalytics | Jun 2022 - Jan 2023

- Optimized the analysis of public Sustainability/Annual Reports by developing an EU Taxonomy Detection tool, reducing manual review time by 25% through the implementation of a RAG-based data model to automate disclosure detection.
- Optimized ESG data acquisition by assisting on the development of the 'Thematic Crawler' to automate the extraction of information from web sources and internal databases, reducing manual research time by 75%
- Enhanced business analysis accuracy by reducing noise by 50% for the Thematic Crawler by engineering a multi-layer text mining framework to streamline data processing
- Input data from the Thematic Crawler was provided by stakeholders, so data cleaning/verification was performed on the input data for each request. This helped prevent erroneous data in the reports.

Business Analyst | Morningstar Sustainalytics | Nov 2021 - Jun 2022

- Orchestrated the deployment of customized Thematic Crawler applications using Python & SQL, resulting in a 75% reduction in manual data extraction time, by facilitating cross-functional requirement-gathering sessions to align technical capabilities with diverse business needs
- Delivered actionable insights via Excel reporting to support strategic decision-making for 5 business units

Technical Skills

- **Languages:** Python, VBA, SQL
- **Frameworks:** NumPy/Pandas, scikit-learn
- **Tools:** LLM, AWS, Docker, Git

Projects

Rogers Centre Attendance Forecast

Loaded attendance data from MLB's StatsAPI into a Postgres database, then created a regression forecast using XGBoost of the attendance figures.

Education

- **B.S. in Statistics**, University of Toronto